Name: ____________________________ Date: ________________ Period: _________

1. **Model from Restrictions.** A farmer has 120 feet of fencing to enclose a rectangular garden. Both pairs of opposite sides require fencing (no wall). Let w = width in feet.

   (a) Write the constraint equation and solve for length ℓ in terms of w.

   Constraint: _____________________________ $\Rightarrow$ $\ell =$ _______________________

   (b) Write the area function $A(w)$.

   $A(w) =$ ____

   (c) State the domain restriction.

   Domain: _____________________________

   (d) Find the width and length that maximize area.

   $w_{\text{max}} =$ _________________ ft $\ell_{\text{max}} =$ _________________ ft $A_{\text{max}} =$ _________________ ft^2

2. **Rational Inverse-Proportion Model.** The gravitational force between two objects is modeled by $F(d) = \dfrac{k}{d^2}$, where d is the distance in meters and $k = 4 \times 10^{14}$ N·m^2.

   (a) State the domain restriction and describe what happens to F as $d \to \infty$.

   Domain: _________________ As $d \to \infty$, $F(d) \to$ ___________.

   (b) Compute $F(2 \times 10^7)$ and $F(4 \times 10^7)$. What is the ratio $F(2 \times 10^7) \,/\, F(4 \times 10^7)$?

   $F(2 \times 10^7) =$ _______________________ N $F(4 \times 10^7) =$ _______________________ N
   Ratio: _________________ This illustrates the _______________________ law.

3. **Apply the Model — Average Rate of Change.** A ball's position (in feet) after t seconds is modeled by $s(t) = -16t^2 + 64t + 5$.

   (a) Evaluate $s(1)$ and $s(3)$.

   $s(1) =$ _______________________ ft $s(3) =$ _______________________ ft

   (b) Compute the average rate of change of s on $[1, 3]$. Include units.

   $\text{ARC} = \dfrac{s(3) - s(1)}{3 - 1} = \dfrac{\rule{5cm}{0.4pt}}{\rule{2cm}{0.4pt}} =$ _________________
   Units: _______________________

   (c) Interpret the ARC: what does it tell you about the ball's motion?

   ____

   ____

4. **Interpret with Units.** A model gives the volume of water $V(t)$ in gallons in a tank t minutes after draining begins. The average rate of change on $[0, 10]$ is -15 gal/min.

(a) Is the tank filling or draining? How do you know from the sign of the ARC?

____

(b) If $V(0) = 200$ gal, estimate $V(10)$ using the average rate of change. Is this an exact value or an approximation? Why?

$V(10) \approx$ ____________ gal. This is an ____________ because ____________

5. **Piecewise Model.** A downtown parking garage charges:

- A flat fee of \$3.00 for the first hour (or any part thereof).
- \$2.00 per hour for each additional hour after the first.

Let $C(t)$ = total cost in dollars for t hours of parking, $t > 0$.

(a) Write the piecewise model. (Treat hours as a continuous variable.)

$$C(t) = \begin{cases} \rule{4cm}{0.4pt} & \text{if } 0 < t \leq 1 \\ \rule{4cm}{0.4pt} & \text{if } t > 1 \end{cases}$$

(b) Compute $C(0.5)$, $C(1)$, $C(2.5)$, and $C(4)$.

$C(0.5) =$ ________ $C(1) =$ ________ $C(2.5) =$ ________ $C(4) =$ ________

(c) Is $C(t)$ a polynomial function? Explain.

____

6. **AP-Style Multiple Choice.** A function model for a company's revenue (in thousands of dollars) is $R(x) = -x^2 + 8x$, where x is the number of units sold (in hundreds), $0 \leq x \leq 8$. What is the average rate of change of R on $[2, 6]$?

(A) 0 **(B)** 4 **(C)** -4 **(D)** 8

Answer: ______ Show work:

____

____

Reflection

What strategy do you use first when deciding whether a model should be polynomial or rational? ____________

____